

Lecture 13

Time Series Analysis II

STAT 8020 Statistical Methods II

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Modeling Trend, Seasonality, and Noise

Recall the trend, seasonality, noise decomposition:

$$Y_t = \mu_t + s_t + \eta_t,$$

where

- μ_t : smooth long-term change in the mean level of the series;
- s_t : seasonal or periodic pattern that repeats over time;
- η_t : stochastic fluctuations that may be correlated across time.

A time series may contain both systematic patterns (trend and seasonality) and random fluctuations (noise)

We are going to learn two approaches for estimating s_t , the **seasonal component**

- Let's consider the situation that a time series consists of seasonal component only (assuming the trend has been estimated/removed), that is,

$$Y_t = s_t + \eta_t,$$

with $\{s_t\}$ having period s (i.e., $s_{t+js} = s_t$ for all integers j and t), $\sum_{t=1}^s s_t = 0$ and $\mathbb{E}(\eta_t) = 0$

- Question:** How can we estimate a repeating seasonal pattern s_t ?
 - Harmonic regression
 - Seasonal mean model

- A harmonic regression model has the form

$$s_t = \sum_{j=1}^k A_j \cos(2\pi f_j t + \phi_j).$$

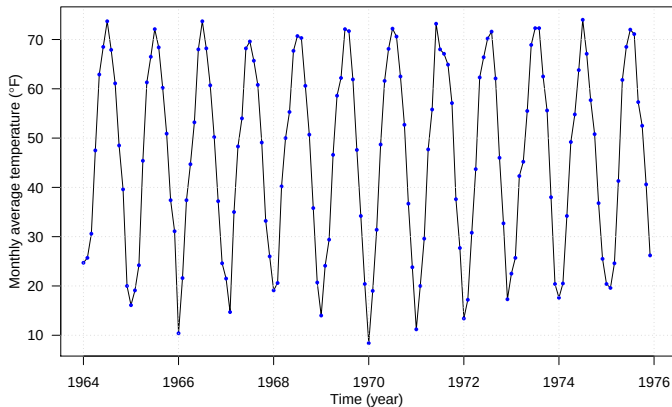
For each $j = 1, \dots, k$:

- $A_j > 0$ is the amplitude of the j -th cosine wave
 - f_j controls the frequency of the j -th cosine wave (how often waves repeats)
 - $\phi_j \in [-\pi, \pi]$ is the phase of the j -th wave (where it starts)
- The above can be expressed as

$$\sum_{j=1}^k \{ \beta_{1j} \cos(2\pi f_j t) + \beta_{2j} \sin(2\pi f_j t) \},$$

where $\beta_{1j} = A_j \cos(\phi_j)$ and $\beta_{2j} = -A_j \sin(\phi_j) \Rightarrow$ **if $\{f_j\}_{j=1}^k$ are known, we can use regression techniques to estimate the parameters $\{\beta_{1j}, \beta_{2j}\}_{j=1}^k$**

Monthly Temperature in Dubuque, IA [Cryer & Chan, 2008]



Let's assume there is no trend in this time series. Here we want to estimate s_t , the seasonal component

Modeling Annual Cycle via Harmonic Regression

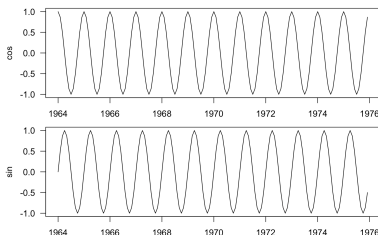
For monthly data:

$$s_t = \beta_0 + \beta_1 \cos\left(\frac{2\pi t}{12}\right) + \beta_2 \sin\left(\frac{2\pi t}{12}\right)$$

The period is 12 months, so one complete cycle occurs every year. In R, harmonic terms are easily generated via

`TSA::harmonic`

```
harmonics <- harmonic(tempdub, 1)
```



⇒ A single pair of sine and cosine terms can capture a smooth annual seasonal cycle

```
```{r}
harReg <- lm(tempdub ~ harmonics)
summary(harReg)
```

Call:

```
lm(formula = tempdub ~ harmonics)
```

Residuals:

Min	1Q	Median	3Q	Max
-11.1580	-2.2756	-0.1457	2.3754	11.2671

Coefficients:

	Estimate	Std. Error	t value	Pr(> t )
(Intercept)	46.2660	0.3088	149.816	< 2e-16 ***
harmonicscos(2*pi*t)	-26.7079	0.4367	-61.154	< 2e-16 ***
harmonicssin(2*pi*t)	-2.1697	0.4367	-4.968	1.93e-06 ***

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

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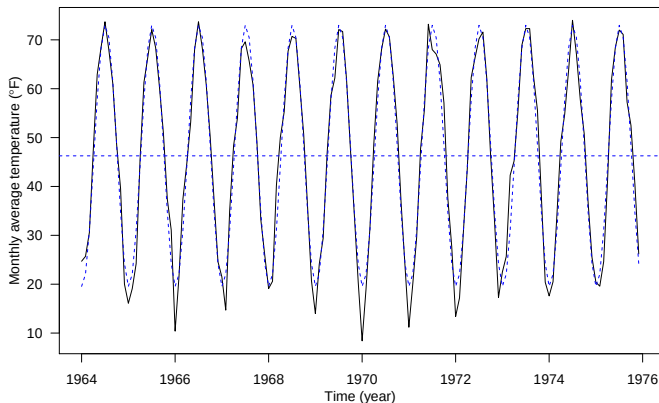
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Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Question: What assumptions are we making here?

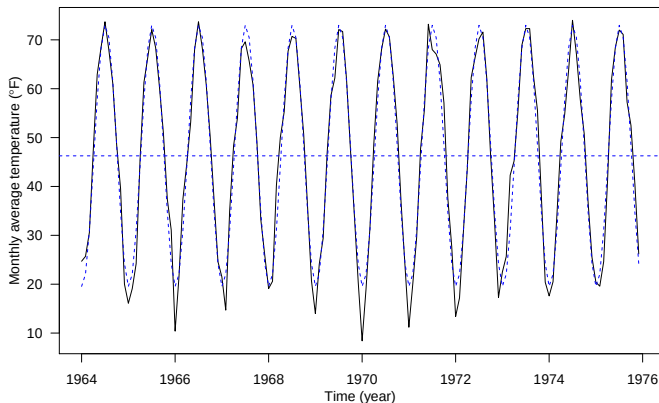
The Harmonic Regression Model Fit



Assumptions:

- Seasonal pattern is sinusoidal
- Amplitude remains constant over time
- Seasonal pattern does not change across years
- Errors are independent with constant variance

The Harmonic Regression Model Fit



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- Errors are independent with constant variance

Question: What can be the model limitations?

Harmonics regression assumes the seasonal pattern has a regular shape $\Rightarrow$ may be too restrictive for complex seasonal patterns

- A less restrictive approach is to model $\{s_t\}$ as

$$s_t = \begin{cases} \beta_1 & \text{for } t = 1, 1 + d, 1 + 2d, \dots & ; \\ \beta_2 & \text{for } t = 2, 2 + d, 2 + 2d, \dots & ; \\ \vdots & \vdots & ; \\ \beta_d & \text{for } t = d, 2d, 3d, \dots & . \end{cases}$$

- A **seasonal means model**: each month has a separate mean, equivalent to an ANOVA with month as a factor

Call:

`lm(formula = tempdub ~ month - 1)`

Residuals:

Min	1Q	Median	3Q	Max
-8.2750	-2.2479	0.1125	1.8896	9.8250

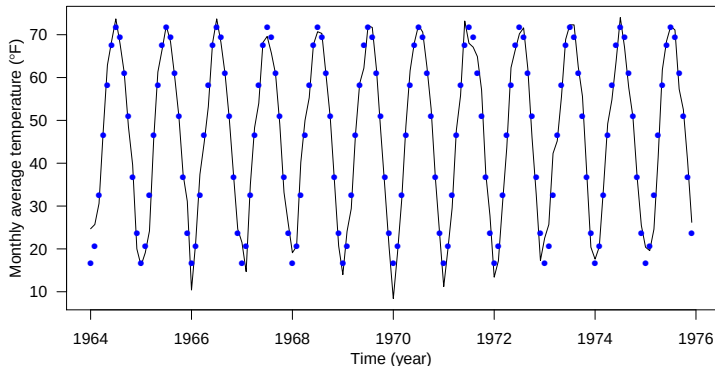
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
monthJanuary	16.608	0.987	16.83	<2e-16	***
monthFebruary	20.650	0.987	20.92	<2e-16	***
monthMarch	32.475	0.987	32.90	<2e-16	***
monthApril	46.525	0.987	47.14	<2e-16	***
monthMay	58.092	0.987	58.86	<2e-16	***
monthJune	67.500	0.987	68.39	<2e-16	***
monthJuly	71.717	0.987	72.66	<2e-16	***
monthAugust	69.333	0.987	70.25	<2e-16	***
monthSeptember	61.025	0.987	61.83	<2e-16	***
monthOctober	50.975	0.987	51.65	<2e-16	***
monthNovember	36.650	0.987	37.13	<2e-16	***
monthDecember	23.642	0.987	23.95	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

MonthJuly = 71.7 $\Rightarrow$ Average July temperature $\approx 71.7^{\circ}\text{F}$

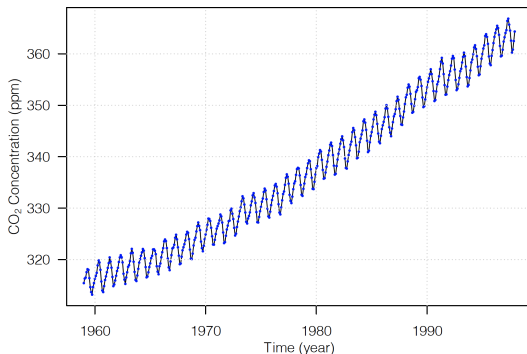
The Seasonal Means Model Fit



Comparison with Harmonic Regression

Seasonal means provide greater flexibility but require more parameters

Estimating the Trend and Seasonal Components Together

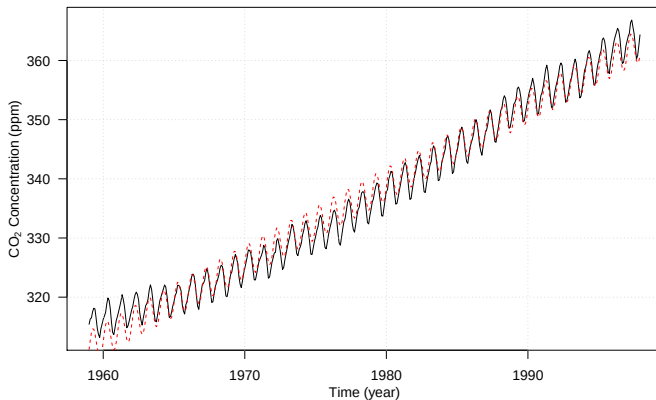


Let's perform a regression analysis to model both μ_t (assuming a linear time trend) and s_t (using $\cos$ and $\sin$)

```
```{r}
time <- as.numeric(time(co2))
harmonics <- harmonic(co2, 1)

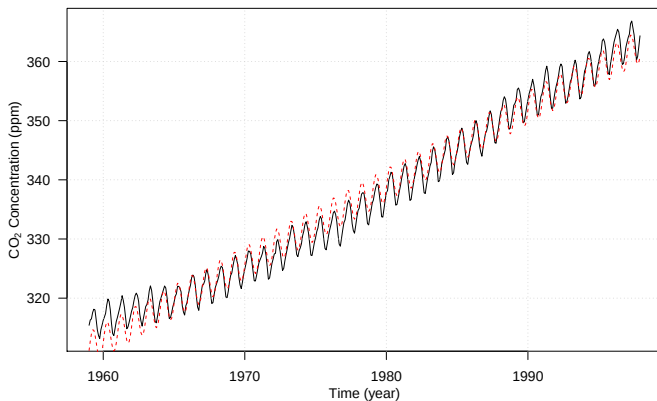
lm_trendSeason <- lm(co2 ~ time + harmonics)
summary(lm_trendSeason)
```

# The Regression Fit



**Question:** How well the model fits the data?

# The Regression Fit



**Question:** How well the model fits the data?

**Observations:**

- Seasonal pattern captured reasonably well
- Trend appears nonlinear

## Backshift Operator and Differencing

- The **backshift operator**  $B$  shifts a series back by one time period:

$$BY_t = Y_{t-1}$$

- More generally,

$$B^q Y_t = Y_{t-q}$$

- Examples:

$$BY_t = Y_{t-1}, \quad B^2 Y_t = Y_{t-2}, \quad B^{12} Y_t = Y_{t-12}.$$

- The **first difference** can be written as

$$\nabla Y_t = Y_t - Y_{t-1} = (1 - B)Y_t.$$

- Higher-order differences are defined recursively:

$$\nabla^q Y_t = \nabla(\nabla^{q-1} Y_t).$$

- A **seasonal difference** compares observations from the same season:

$$Y_t - Y_{t-s} = (1 - B^s)Y_t$$

## Why Do We Difference a Time Series?

- Many time series exhibit **trend** and/or **seasonality**
- ARMA models typically require the series to be approximately **stationary**
- First differencing removes trend:

$$\nabla Y_t = Y_t - Y_{t-1}$$

- Seasonal differencing removes seasonality:

$$(1 - B^s)Y_t = Y_t - Y_{t-s}$$

### Key Takeaway

Differencing is often used to transform a nonstationary series into one that can be modeled by an ARMA process

## Formal Definition of a SARIMA Model

Let  $d$  and  $D$  be non-negative integers. Then  $\{X_t\}$  is a **seasonal ARIMA**  $(p, d, q) \times (P, D, Q)$  **process with period  $s$**  if

$$Y_t = \nabla^d \nabla_s^D X_t = (1 - B)^d (1 - B^s)^D X_t,$$

is an ARMA process define by

$$\phi(B)\Phi(B^s)Y_t = \theta(B)\Theta(B^s)Z_t,$$

where  $\{Z_t\} \sim \text{WN}(0, \sigma^2)$ .

### Interpretation

SARIMA extends ARMA by incorporating ordinary and seasonal differencing, together with nonseasonal and seasonal AR and MA components

Consider a monthly time series  $\{X_t\}$  with both a trend, and a seasonal component of period  $s = 12$ .

- Suppose we know the values of  $d$  and  $D$  such that  $Y_t = (1 - B)^d(1 - B^{12})^D X_t$  is **stationary**
- We can arrange the data this way:

	Month 1	Month 2	...	Month 12
Year 1	$Y_1$	$Y_2$	...	$Y_{12}$
Year 2	$Y_{13}$	$Y_{14}$	...	$Y_{24}$
$\vdots$	$\vdots$	$\vdots$	...	$\vdots$
Year $r$	$Y_{1+12(r-1)}$	$Y_{2+12(r-1)}$	...	$Y_{12+12(r-1)}$

Here we view each column (month) of the data table from the previous slide as a **separate time series**

- For each month  $m$ , we assume the same ARMA( $P, Q$ ) model. We have

$$\begin{aligned} Y_{m+12s} - \sum_{i=1}^P \Phi_i Y_{m+12(s-i)} \\ = U_{m+12s} + \sum_{j=1}^Q \Phi_j U_{m+12(s-j)}, \end{aligned}$$

for each year  $s = 0, \dots, r-1$ , where

$\{U_{m+12s:s=0,\dots,r-1}\} \sim \text{WN}(0, \sigma_U^2)$  for each  $m$

- We can write this as

$$\Phi(B^{12})Y_t = \Theta(B^{12})U_t,$$

and this defines the **inter-annual model**

## The Intra-Annual Model

After removing the trend, seasonality, and inter-annual effects, the series may still exhibit short-term dependence across months

We model the remaining dependence using an  $\text{ARMA}(p, q)$  process:

$$\phi(B)U_t = \theta(B)Z_t,$$

where

$$Z_t \sim WN(0, \sigma^2).$$

### SARIMA Key Idea

Annual and monthly differencing remove trends, the annual ARMA model captures inter-year dependence, and the ARMA model captures the remaining month-to-month correlation

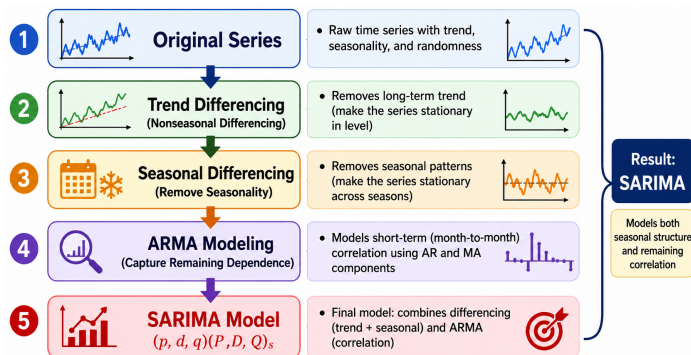
1. Transform the data if necessary (e.g., logarithm for variance stabilization)
2. Apply regular and seasonal differencing until the series is approximately **stationary**

$$Y_t = (1 - B)^d (1 - B^s)^D X_t$$

3. Use the ACF/PACF at seasonal lags ( $s, 2s, 3s, \dots$ ) to identify  $P$  and  $Q$
4. Use the ACF/PACF at nonseasonal lags ( $1, \dots, s - 1$ ) to identify  $p$  and  $q$

## Modeling SARIMA Processes (Cont'd)

5. Use **maximum likelihood method** to fit the models
6. Use model summaries, diagnostics, AIC to determine the best SARIMA model
7. Conduct forecast



## Two Types of Dependence in SARIMA

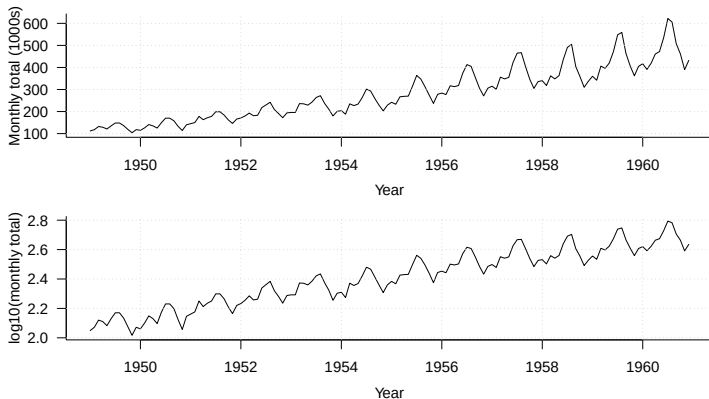
Component	Captures	Example Lag
$AR(p)$	Month-to-month dependence	1,2,3
$MA(q)$	Short-term shocks	1,2,3
$SAR(P)$	Seasonal dependence	12,24,36
$SMA(Q)$	Seasonal shocks	12,24,36

### Interpretation

SARIMA combines short-term dependence and seasonal dependence in a single model

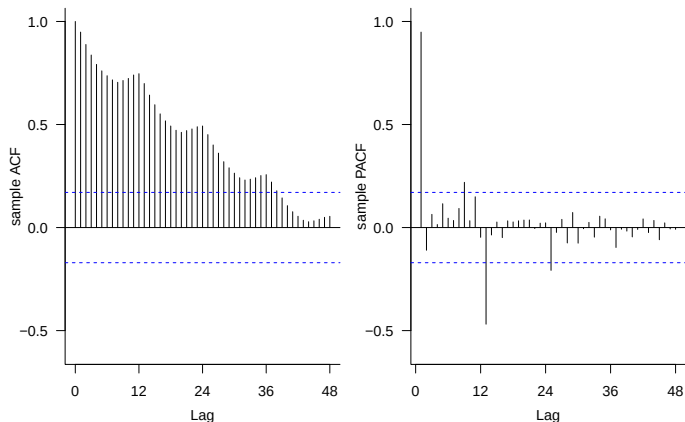
## Airline Passengers Example [Box and Jenkins, 1970]

The `AirPassengers` data record monthly international airline passenger totals from 1949–1960



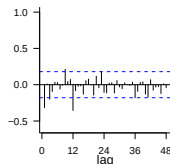
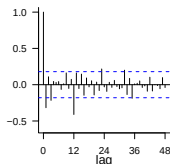
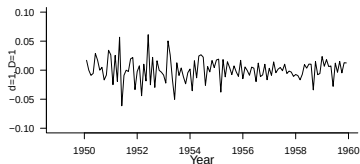
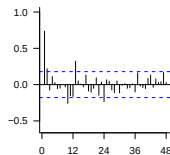
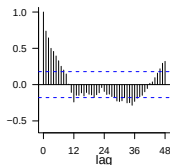
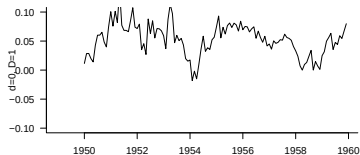
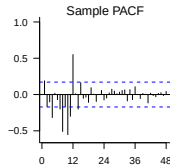
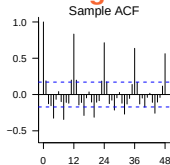
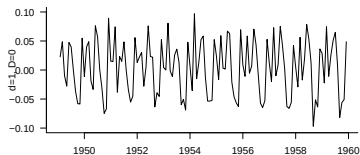
The seasonal fluctuations increase as the number of passengers grows, indicating nonconstant variance. A logarithmic transformation is therefore appropriate

## Sample ACF/PACF Plots



- A slowly decaying ACF and a PACF spike near one at lag 1 suggest nonstationarity and the need for regular differencing
- Strong spikes at seasonal lags (12, 24, 36) indicate annual seasonality and suggest seasonal differencing

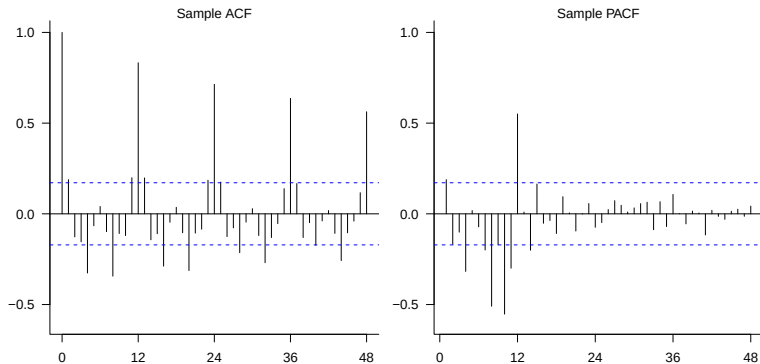
# Trying Different Orders of Differencing



Apply the smallest amount of differencing needed to remove trend

## Choosing Candidate SARIMA Models

The first difference  $d = 1$  appears sufficient to remove the trend. The ACF/PACF will now be used to identify the seasonal and nonseasonal AR/MA orders



Now we need to choose  $P$ ,  $Q$ ,  $p$ , and  $q$

## Fitting a SARIMA(1, 1, 0) $\times$ (1, 0, 0) Model

```
> fit1 <- arima(diff.1.0, order = c(1, 0, 0), seasonal = list(order = c(1, 0, 0), period = 12))
> fit1
```

Call:

```
arima(x = diff.1.0, order = c(1, 0, 0), seasonal = list(order = c(1, 0, 0),
 period = 12))
```

Coefficients:

	ar1	sar1	intercept
	-0.2667	0.9291	0.0039
s.e.	0.0865	0.0235	0.0096

sigma^2 estimated as 0.0003298: log likelihood = 327.27, aic = -646.54

### Estimated Model

Both the nonseasonal AR(1) and seasonal AR(1) terms are statistically significant.

## Diagnostic Check: SARIMA(1, 1, 0) $\times$ (1, 0, 0)<sub>12</sub>

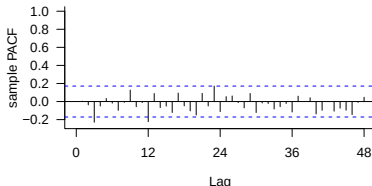
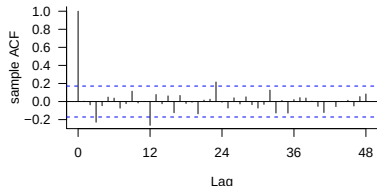
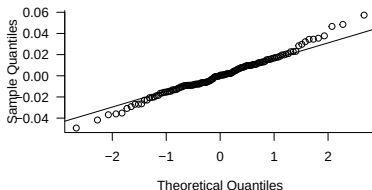
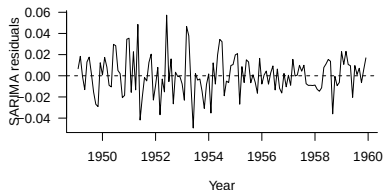
```
> Box.test(fit1$residuals, lag = 48, type = "Ljung-Box")
```

Box-Ljung test

data: fit1\$residuals

X-squared = 55.372, df = 48, p-value = 0.2164

Ljung-Box test examines residual independence



- The spread of the residuals is larger in 1949-1955 compared to the later years and the residual distribution has heavy tails
- The Ljung-Box test result indicates the fitted SARIMA  $(1, 1, 0) \times (1, 0, 0)$  has sufficiently account for the temporal dependence
- 95% CI for  $\phi_1$  and  $\Phi_1$  do not contain zero  $\Rightarrow$  no need to go with simpler model

Our estimated model is

$$(1 + 0.2667B)(1 - 0.9291B^{12})(X_t - 0.0039) = Z_t,$$

where  $\{Z_t\} \stackrel{i.i.d.}{\sim} N(0, \sigma^2)$  with  $\hat{\sigma}^2 = 0.00033$

## Comparing with a SARIMA(0,1,0) $\times$ (1,0,0) Model

```
> (fit2 <- arima(diff.1.0, seasonal = list(order = c(1, 0, 0), period = 12)))
```

Call:

```
arima(x = diff.1.0, seasonal = list(order = c(1, 0, 0), period = 12))
```

Coefficients:

	sar1	intercept
	0.9081	0.0040
s.e.	0.0278	0.0108

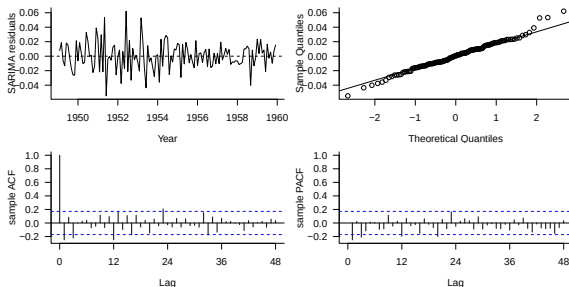
sigma^2 estimated as 0.0003616: log likelihood = 322.75, aic = -639.51

```
> Box.test(fit2$residuals, lag = 48, type = "Ljung-Box")
```

Box-Ljung test

data: fit2\$residuals

X-squared = 80.641, df = 48, p-value = 0.002209

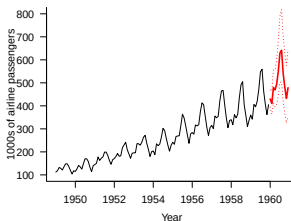
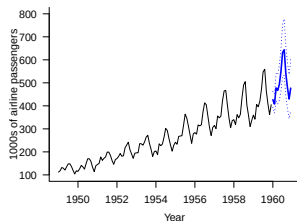
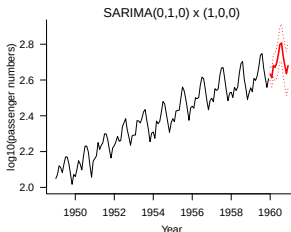
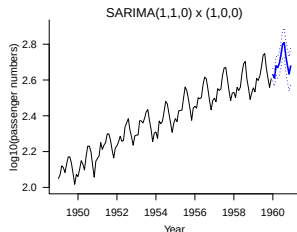


Here we drop the AR(1) term

- The residual plots looks quite similar to before: The spread of the residuals is larger in 1949-1955 compared to the later years and the residual distribution has heavy tails
- Both  $\hat{\sigma}^2$  and AIC increase (compared with model fit1)
- The lag 1 of ACF and PACF now lies outside the IID noise bounds. The Ljung-Box  $p$ -value of 0.0022, leads us to reject the IID residual assumption

In conclusion, the SARIMA(1, 1, 0)  $\times$  (1, 0, 0) model fits better than SARIMA(0, 1, 0)  $\times$  (1, 0, 0)

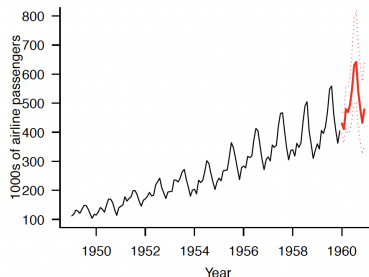
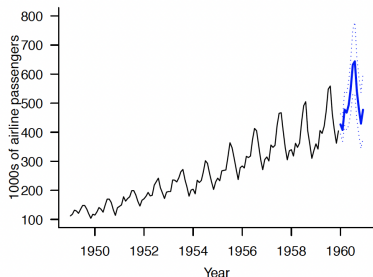
# Forecasting the 1960 Data



Good forecasting performance requires

- 1 Accurate point forecasts
- 2 Reliable prediction intervals

# Evaluating Forecast Performance



Metrics	Model Fit1	Model Fit2
Root Mean Square Error	30.36	31.32
Mean Relative Error	0.057	0.060
Empirical Coverage	0.917	1.000

These slides cover two methods for estimating **seasonality**:

- **Harmonic regression models**
- **Seasonal ARIMA Models**
- Ways to evaluate **forecasting** performance

R functions to know:

- `harmonic` (under the package `TSA`) for constructing harmonic functions
- Incorporating `seasonal = list(order = c(P, D, Q), period = s)` in `arima` for SARIMA modeling